

CONTAINER-ADMISSIBILITY OF Vec IS NOT A THEOREM OF ZFC

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ABSTRACT. A base category $\mathcal{C}$ is $\triangleleft$ -admissible when the family of endofunctors it presents through the container denotation $\llbracket - \rrbracket$ is closed under composition—the minimal condition under which the composition of directed containers has a denotational meaning over $\mathcal{C}$. We show that $\triangleleft$ -admissibility of the category Vec_k of vector spaces over a countable field is *not provable in ZFC*. The governing question (Conjecture 6.2 / Gap- S) reduces to the projectivity, over the full linear ring $S = \text{End}_k(E)$ of a countable-dimensional space E , of the countable product $M = \prod_{\mathbb{N}} (\bigoplus_{\mathbb{N}} E)$ of a non-finitely-generated projective. We first collapse the conjecture to a Dual-Baer test— M is projective if and only if it is R -projective—then prove, by a Baer–Specker cardinal count, that M projective forces $2^{\aleph_0} = 2^{\aleph_1}$. Hence under the Continuum Hypothesis M is not projective, Vec is $\triangleleft$ -inadmissible, and the conjecture fails; so it is not a theorem of ZFC. In the residual case $2^{\aleph_0} = 2^{\aleph_1}$ the count is silent, and we reframe the whole question through a single homological invariant: the conjecture is equivalent to $\text{pd}_S M = 0$, and the cokernel of M inside a free module is the countable product of a projective-dimension-one seed, so that admissibility becomes an instance of the Osofsky–Jensen problem of whether a product raises projective dimension. This residual is genuinely open—both a ZFC refutation and full independence are live—though the structural evidence (a large dual, a collapse threshold that forcing can arrange while preserving reflection, and a Whitehead-shaped decomposition) leans toward independence.

1. INTRODUCTION

The problem. Directed containers are the polynomial-functor presentation of small categories: under the equivalence $\text{DCont} \simeq \text{Cat}$ a directed container *is* a small category, its shapes the objects and its positions the morphisms out of an object [1, 16]. This is the structural backbone of a program that models stateful and interactive systems—agent orchestration, supply networks, tool-calling protocols—as directed containers and reads their compositional correctness off categorical invariants. The engine of that program is a composition operation: containers compose by *substitution*, written $\triangleleft$, which on the underlying polynomial functors is composition of functors.

For containers valued in Set this is unproblematic; substitution always exists. But the program’s ambition is to run the same machinery over other bases— Vec , for tax and blockchain arithmetic; additive categories, for the linear systems of machine learning—and there the first question is whether substitution has a meaning at all. Following the weakest useful formulation, call a base category $\mathcal{C}$ $\triangleleft$ -admissible if the image of the container denotation functor $\llbracket - \rrbracket : \text{Fam}(\mathcal{C}^{\text{op}}) \rightarrow [\mathcal{C}, \mathcal{C}]$ is closed under composition: for all containers p, q there is a container $p \triangleleft q$ with $\llbracket p \triangleleft q \rrbracket \cong \llbracket p \rrbracket \circ \llbracket q \rrbracket$ (Definition 2.1). Admissibility is the license to compose. Without it, there is no denotational composition of directed containers over $\mathcal{C}$, and the whole apparatus stalls at the base.

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The gap. Set is admissible and Vec sits at the opposite degeneracy: its monoidal unit for $\triangleleft$ is the zero object, which makes the shape data of a composite invisible and turns admissibility into a delicate closure question rather than a triviality [8]. That question was isolated as a single conjecture about additive endofunctors of Vec —Conjecture 6.2, also called *Gap- S* —and had stood open through a sequence of failed attempts to construct an “exotic” obstruction by hand. The September 2026 *module reduction* [9] recast it as a concrete problem in the module theory of the full linear ring $S = \text{End}_k(E)$ but stopped short of a decision, and—this is the point of departure for the present note—it explicitly recorded that a natural cardinality obstruction “cannot refute projectivity.” It can.

The result. Fix a countable field k , let $E = k^{(\mathbb{N})}$ (countable dimension), $S = \text{End}_k(E)$, let $P = \bigoplus_{\mathbb{N}} E$ be the standard non-finitely-generated projective S -module, and set

$$M = \prod_{\mathbb{N}} P = \prod_{r \in \mathbb{N}} \left(\bigoplus_{m \in \mathbb{N}} E \right).$$

The reduction identifies $\triangleleft$ -admissibility of Vec (in the relevant instance, Conjecture 6.2) with the projectivity of M over S (Theorem 2.3). Our main theorem is a cardinal-counting obstruction to that projectivity.

Theorem 1.1 (Main Theorem, ZFC). *If M is projective as a left S -module, then $2^{\aleph_0} = 2^{\aleph_1}$. Consequently, whenever $2^{\aleph_0} < 2^{\aleph_1}$ —in particular under the Continuum Hypothesis— M is not projective, so Vec is $\triangleleft$ -inadmissible and Conjecture 6.2 is false. Hence Conjecture 6.2 is not a theorem of ZFC.*

This is, to our knowledge, the first point at which the compositional-systems program meets genuine set-theoretic sensitivity: a purely categorical closure property of Vec is shown to fail in every model of CH, and CH is consistent, so no ZFC proof of it can exist. (Whether it *also* holds in some model of ZFC—i.e. whether the conjecture is fully independent rather than merely unprovable—is the open question of Section 6.) The reduction is a theorem of ZFC (Section 3); the counting is a theorem of ZFC (Section 4); only the ambient value of the continuum function is variable.

The method. The headline rests on the module reduction (Theorem 2.3, which identifies the conjecture with “ M projective”) together with a single new idea: a *Baer–Specker cardinal count*. If M were projective, Kaplansky’s theorem [7] decomposes it as a direct sum $\bigoplus_{i \in I} M_i$ of countably generated pieces; because M is not countably generated the index set has $|I| \geq \aleph_1$; each summand has nonzero dual into E , so the dual $\text{Hom}_S(M, E)$ has k -dimension at least $2^{|I|} \geq 2^{\aleph_1}$; but a slenderness computation pins that dimension at exactly $2^{\aleph_0}$. The collision $2^{\aleph_1} \leq 2^{\aleph_0}$ is the theorem.

Alongside it we record an independent structural fact of its own interest—a *Dual-Baer collapse* (Theorem 3.4): S contains a free left ideal of rank $\mathfrak{c}$ (a Baer–Specker construction), and this is exactly enough, via the projectivity-domain calculus behind Trlifaj’s Dual Baer Criterion [17], to force every R -projective left S -module of at most $\mathfrak{c}$ generators to be projective. Since $|M| \leq \mathfrak{c}$, the conjecture is equivalent not only to “ M projective” but to the single lifting test “ M R -projective,” and—a point we return to in Section 6—this closes the most obvious independence shortcut, since the packaged Dual-Baer independence theorem requires a semiartinian ring and S is not one. The collapse is not a link in the proof of the Main Theorem; the count settles that directly.

The delicate feature of the count—and the reason the verdict is conditional rather than absolute—is that it is precisely the argument that proves the Baer–Specker group $\mathbb{Z}^{\mathbb{N}}$ non-free over $\mathbb{Z}$, but with one term changed. Over $\mathbb{Z}$ the dual $\text{Hom}(\mathbb{Z}^{\mathbb{N}}, \mathbb{Z})$ collapses to countable rank

(Specker’s theorem), producing an *absolute* $2^{\mathfrak{c}}$ -versus- $\aleph_0$ contradiction. Over S the identity $\prod_{\mathbb{N}} E \cong S$ keeps the dual *large*, of dimension $\mathfrak{c}$; the very freeness that voided every naive obstruction is what demotes the contradiction to the conditional gap $2^{\aleph_1}$ versus $2^{\aleph_0}$.

Consequences. Theorem 1.1 settles the standing working hypothesis of the program—that $\text{Gap-}S$ was “plausibly true / ZFC-decidable”—in the negative: the composition of directed containers over Vec is not unconditionally well defined. It also corrects a specific misstep in [9], which derived the inequality $2^{|I|} \leq 2^{\aleph_0}$ and dismissed it; the missing observation is $|I| \geq \aleph_1$, which converts it into $2^{\aleph_1} \leq 2^{\aleph_0}$. Structurally (Section 5) M turns out to be flat, Mittag-Leffler, and $\aleph_1$ -projective—the exact profile of $\mathbb{Z}^{\mathbb{N}}$ —so that the conjecture is equivalent to M being a direct sum of countably generated submodules, and its failure is a *global* defect orthogonal to local projectivity. In the residual regime $2^{\aleph_0} = 2^{\aleph_1}$ (Sections 5–6) the count is silent, and we reframe the residual as a single homological invariant: the conjecture is equivalent to $\text{pd}_S M = 0$, where the cokernel of M inside the free module $\Phi \cong S$ is the countable product $\prod_{\mathbb{N}} N_0$ of a projective-dimension-one seed $N_0 = \prod_{\mathbb{N}} E / \bigoplus_{\mathbb{N}} E$ (Theorem 5.4). Admissibility thus becomes the Osofsky–Jensen question of whether a countable product of pd-1 modules raises projective dimension over S . We bound $\text{pd}_S M$ between 1 (when $2^{\aleph_0} < 2^{\aleph_1}$, the Main Theorem) and a continuum-degree $d + 1$ (a Gruson–Jensen filtration bound), and record that *both* a ZFC refutation and full independence remain live, with an evidential—not proved—lean toward independence and the sharp open question (*can* $\text{pd}_S M = 0$?) that would decide it.

Outline. Section 2 fixes the container framing and states the module reduction. Section 3 proves the Dual-Baer collapse (Theorem 3.4). Section 4 proves the Main Theorem. Section 5 places M in the almost-free hierarchy, identifies the controlling cokernel as $\prod_{\mathbb{N}} N_0$, derives the projective-dimension reframing $\text{Conj. 2.2} \Leftrightarrow \text{pd}_S M = 0$ with its bounds, and records the (order-free, continuum-independent) κ -projectivity reduction. Section 6 treats the residual case and the independence lean. Section 7 draws the consequences for the container program.

Provenance and honesty. Every claim graded “ZFC” below is a proved theorem, verified adversarially in the source PROVE sessions [9, 10, 11, 12, 13]. The proved content (Sections 2–5) rests on the classical literature and on two deep-read external sources: Trlifaj’s Dual Baer Criterion [17] and the Herbera–Trlifaj almost-free theory [6], both read at source. The classical Gruson–Jensen estimate— $\text{pd} \leq n + 1$ for an $\aleph_n$ -generated flat module [5]—is invoked at its standard statement. The lean toward independence in Section 6 is *evidential and explicitly not a proof*; the Herbera–Trlifaj results it quotes (Theorem 2.9(i), Example 6.8, Corollary 7.3) are used only to locate the open problem, never to support a proved claim. One sharpening—that Osofsky’s continuum-sensitive global-dimension theorem [14] would give $\text{pd}_S M \leq d$, hence $\text{pd}_S M = 1$ under CH—rests on a recollection of that theorem’s exact value that we have *not* verified at source; it appears only in Remark 5.7, and no proved claim depends on it.

2. THE CONTAINER FRAMING AND THE MODULE REDUCTION

We recall only what the proof uses. For containers, directed containers, and the equivalence with Cat , see [1, 16]; for the admissibility trichotomy over general bases, see [8].

2.1. Containers and admissibility. Let $\mathcal{C}$ be a category with the small coproducts and products used below. Write $\text{Fam}(\mathcal{C}^{\text{op}})$ for the category whose objects are pairs (S, P) with S a set and $P: S \rightarrow \mathcal{C}$ a family; its *denotation* is the endofunctor

$$\llbracket S, P \rrbracket X = \prod_{s \in S} \mathcal{C}(P(s), X).$$

When $\mathcal{C} = \text{Set}$ this is the usual container functor $X \mapsto \coprod_s X^{P(s)}$; over an additive $\mathcal{C}$ the internal $\text{hom } \mathcal{C}(P(s), -)$ replaces the power.

Definition 2.1. $\mathcal{C}$ is $\triangleleft$ -admissible if for all $p, q \in \text{Fam}(\mathcal{C}^{\text{op}})$ there exists $p \triangleleft q \in \text{Fam}(\mathcal{C}^{\text{op}})$ with $\llbracket p \triangleleft q \rrbracket \cong \llbracket p \rrbracket \circ \llbracket q \rrbracket$; equivalently, if the image of $\llbracket - \rrbracket$ is closed under composition of endofunctors.

This is deliberately the weakest hypothesis—no monoidal structure, no canonical choice of $p \triangleleft q$ —so that a non-existence result proved against it is maximally strong. Over Set every category is admissible; over Vec admissibility is subtle because the unit object is 0 and the shape set of a composite becomes invisible, which is where the following single instance concentrates the difficulty.

2.2. The additive functors over Vec . Work in $\mathcal{A} = \text{Add}(\text{Vec}_k, \text{Vec}_k)$, the category of additive endofunctors of Vec_k and natural transformations. Write $h_k = \text{Hom}_k(k, -) \cong \text{id}$ for the identity (representable at k) and $h_E = \text{Hom}_k(E, -)$ for the representable at E . The relevant functors are

$$A = \bigoplus_{\mathbb{N}} h_k \quad (A(X) = \bigoplus_m X), \quad F = \prod_{\mathbb{N}} A \quad (F(X) = \prod_r \bigoplus_m X),$$

together with $F_{\text{fin}} = \bigoplus_{\mathbb{N}} A$ and the quotient $Q = F/F_{\text{fin}}$. Since $E = k^{(\mathbb{N})}$, one has $h_E = \text{Hom}_k(k^{(\mathbb{N})}, -) = \prod_{\mathbb{N}}(-)$, hence $F = \text{Hom}_k(E, \bigoplus_{\mathbb{N}} -)$: the functor of the conjecture.

Conjecture 2.2 (6.2 / Gap-S). *F is projective in $\mathcal{A}$ (equivalently, $F \cong \bigoplus_{\Lambda} h_E$ for some Λ). Equivalently, the comparison map $\iota_*: \text{Ext}_{\mathcal{A}}^1(F, F_{\text{fin}}) \rightarrow \text{Ext}_{\mathcal{A}}^1(F, F)$ is injective.*

The reduction of $\triangleleft$ -admissibility of Vec to Conjecture 2.2 is established in [8, 9]: the substitution of the two containers presenting $\bigoplus_{\mathbb{N}}$ produces exactly F , and admissibility at that instance asks that F be again a coproduct of representables—the projective objects of $\mathcal{A}$. We take this reduction as given and pass to modules.

2.3. The module reduction. Let $S = \text{End}_k(E)$, the ring of column-finite $\mathbb{N} \times \mathbb{N}$ matrices over k , and let $\mathcal{C}_0 \subseteq \text{Vec}_k$ be the full subcategory of spaces of dimension $\leq \aleph_0$.

Theorem 2.3 (Module reduction; [9, Thm. 1–2]). *Every $X \in \mathcal{C}_0$ is a retract of E , and $E \oplus E \cong E$; hence $\mathcal{C}_0 \simeq \text{Kar}(S)$ and $\text{Add}(\mathcal{C}_0, \text{Vec}) \simeq S\text{-Mod}$ via $T \mapsto T(E)$. The functors A, F, F_{fin}, Q are $\aleph_1$ -continuous (each value has countable support), so restriction to $\mathcal{C}_0$ is faithful and*

$$\text{Nat}_{\text{Vec}}(T, T') \cong \text{Hom}_S(T(E), T'(E)),$$

naturally in T, T' , and $\text{Ext}_{\mathcal{A}}^1(T, T') \cong \text{Ext}_S^1(T(E), T'(E))$ for $T, T' \in \{A, F, F_{\text{fin}}, Q, \dots\}$. Under $T \mapsto T(E)$,

functor	left S -module
$\text{id} = h_k$	E (simple projective = Se_{00})
$A = \bigoplus_m h_k$	$P = \bigoplus_m E$ (projective, countably generated, non-f.g.)
h_E	S (free rank 1)
$F = \prod_r A$	$M = \prod_r P$
$F_{\text{fin}} = \bigoplus_r A$	$M_{\text{fin}} = \bigoplus_r P$
$Q = F/F_{\text{fin}}$	M/M_{fin}

In particular Vec is $\triangleleft$ -admissible (Conjecture 2.2) if and only if M is projective over S .

We use freely the following facts about S , all proved in [9, §2–3]. For countable k : $|S| = |k|^{\aleph_0} = \mathfrak{c}$ and $\dim_k S = \mathfrak{c}$, so $|M| \leq \mathfrak{c}$.

Proposition 2.4 (Ring facts; [9, Thm. 3]). *$S = \text{End}_k(E)$ is von Neumann regular; hence every S -module is flat and every short exact sequence of S -modules is pure. S is right self-injective but not left self-injective; equivalently E is not an injective left S -module. The two-sided ideal lattice is $\{0, F_0, S\}$ with F_0 the finite-rank operators, $\text{Soc}({}_S S) = F_0$, and S/F_0 simple with zero socle—so S is not semiartinian.*

The single most important structural fact is the following freeness identity, the module shadow of the representability of $\prod_{\mathbb{N}} \text{id} = h_E$.

Theorem 2.5 (Freeness; [9, Thm. 2]). *As left S -modules $\prod_{r \in \mathbb{N}} E \cong S$, free of rank 1, via $\Psi(s) = (s \cdot e_r)_r$ (a column-finite matrix is exactly an arbitrary sequence of its columns). Consequently $\Phi := \prod_{r,m} E \cong S$ and $\prod_{\mathbb{N}} S \cong S$, and M sits inside a free module, $M_{\text{fin}} \subseteq \bigoplus_m S \subseteq M \subseteq \Phi \cong S$.*

Corollary 2.6 (Baer–Specker void; [9, Thm. 4]). *The Specker-type identity $\text{Hom}_S(\prod_r E, E) = \bigoplus_r \text{Hom}_S(E, E)$ holds and is consistent with $\prod_r E$ being free (indeed it is free, Theorem 2.5). Hence no “product $\neq$ coproduct” argument can exhibit an exotic non-lifting map or prove M non-projective by a size gap alone.*

Corollary 2.6 explains why every attempt to build an obstruction by hand failed: the mechanism such constructions invoke is neutralised by product-absorption. Theorem 1.1 succeeds precisely because it does not build a map—it counts. Chase’s theorem [3] sharpens the puzzle: all products of projective left S -modules are projective if and only if S is left perfect and right coherent, and S is right coherent (von Neumann regular) but not left perfect; so *some* product of projectives over S is non-projective. Chase’s witness, however, is an *uncountable* power, while $\prod_{\mathbb{N}} S \cong S$ is free—the countable product M falls in exactly the gap Chase leaves open.

Two further facts about the fixed module M are load-bearing below. Both are theorems of ZFC.

Lemma 2.7 (Non-countable generation; [9, §5, F1]). *$M = \prod_r P$ is not countably generated as a left S -module.*

The proof is a diagonalisation against any countable family (a Bergman-type argument in the reduced structure M/M_{fin}); note that cardinality alone cannot see this, since $|M| = \mathfrak{c}$ equals the size of a countably generated S -module. The detailed argument is in [9, §5].

Lemma 2.8 (Small dual; [9, Thm. F(a)]). *$\dim_k \text{Hom}_S(M, E) = 2^{\aleph_0}$. Indeed $\text{Hom}_S(M, E) \cong \bigoplus_r \text{Hom}_S(P, E) \cong \bigoplus_r k^{\mathbb{N}}$ by slenderness of the finite-dimensional target E , and $\dim_k(\bigoplus_r k^{\mathbb{N}}) = \aleph_0 \cdot \mathfrak{c} = 2^{\aleph_0}$.*

Only the upper bound $\dim_k \text{Hom}_S(M, E) \leq 2^{\aleph_0}$ is used in Section 4.

3. THE DUAL-BAER COLLAPSE

Before counting, we record why the two things being counted— M projective and M R -projective—coincide over S . Recall the Dual Baer notion [17]: a left module N is *R -projective* if every homomorphism $N \rightarrow S/I$ into a cyclic quotient by a left ideal I lifts through $S \twoheadrightarrow S/I$. Projective always implies R -projective; the converse is the Dual Baer Criterion, which fails for general rings but holds under a generation bound provided the ring has a large free ideal.

Lemma 3.1 (A free left ideal of rank $\mathfrak{c}$). *S contains a free left ideal $L \cong S^{(\mathfrak{c})}$.*

Proof. By Theorem 2.5, $S \cong \prod_{n \in \mathbb{N}} S$ as left modules. Fix an almost-disjoint family $\{A_i\}_{i \in X}$ of infinite subsets of $\mathbb{N}$ with $|X| = \mathfrak{c}$ (e.g. the branches of $2^{<\omega}$). Define $w_i \in \prod_n S$ by $(w_i)_n = 1_S$ if $n \in A_i$ and 0 otherwise. If a finite left-combination $\sum_k s_{i_k} w_{i_k} = 0$, then for each n , $\sum_{k: n \in A_{i_k}} s_{i_k} = 0$; by almost-disjointness choose $n \in A_{i_{k_0}}$ lying in no other A_{i_k} , forcing $s_{i_{k_0}} = 0$.

Thus $\{w_i\}$ is left-independent and $\bigoplus_i Sw_i \cong S^{(\mathfrak{c})}$; transporting through the isomorphism gives a free left ideal $L \subseteq S$. $\square$

Remark 3.2. This is genuinely stronger than the free ideal of rank $\dim_k E = \aleph_0$ furnished by Trlifaj's Example 2.6 (a partition of a basis), and it is what carries the collapse all the way to $\mathfrak{c} \geq \mu(M)$. The construction is the Baer–Specker almost-disjoint trick used on the left.

Lemma 3.3 (Left Dual Baer upgrade). *Every R -projective left S -module N generated by at most $\mathfrak{c}$ elements is projective.*

Proof. The projectivity domain $\text{Pr}^{-1}(N) = \{X : N \text{ is } X\text{-projective}\}$ is closed under submodules and direct sums [2, 16.12], [17, Lem. 2.2]. If N is R -projective then $S \in \text{Pr}^{-1}(N)$, hence the free ideal $L \cong S^{(\mathfrak{c})}$ of Lemma 3.1 lies in $\text{Pr}^{-1}(N)$, so N is $S^{(\mathfrak{c})}$ -projective. As N is generated by at most $\mathfrak{c}$ elements there is an epimorphism $g: S^{(\mathfrak{c})} \twoheadrightarrow N$; $S^{(\mathfrak{c})}$ -projectivity makes $\text{Hom}(N, -)$ exact on $0 \rightarrow \ker g \rightarrow S^{(\mathfrak{c})} \rightarrow N \rightarrow 0$, so id_N lifts, g splits, and N is a summand of a free module. $\square$

Theorem 3.4 (The collapse, ZFC). *Conjecture 2.2 $\iff M$ is projective $\iff M$ is R -projective.*

Proof. The first equivalence is Theorem 2.3. For the second: projective implies R -projective always; conversely apply Lemma 3.3 to $N = M$, valid because M is generated by at most $|M| \leq \mathfrak{c}$ elements. $\square$

Theorem 3.4 upgrades the conjecture from “ R -projectivity is only necessary” to a genuine ZFC equivalence, and it forecloses the shortest independence route: Trlifaj's independence theorem for the Dual Baer Criterion [17, Thm. 4.4] requires the ring to be *small semiartinian*, which S is not (Proposition 2.4). So the packaged independence theorem does not apply to S ; whatever the status of the conjecture, it is not an instance of the known Dual-Baer independence results. We turn to what does decide it under CH.

4. THE MAIN THEOREM: A BAER–SPECKER CARDINAL COUNT

Lemma 4.1 (Nonzero dual on projectives). *Every nonzero countably generated projective left S -module N has $\text{Hom}_S(N, E) \neq 0$.*

Proof. Fix an embedding $N \subseteq S^{(\mathbb{N})}$ as a direct summand and a nonzero $n = \sum_i n_i e_i \in N$ (finite sum, $n_i \in S$) with some $n_j \neq 0$. As $n_j \in \text{End}_k(E)$ is a nonzero endomorphism, choose $w \in E$ with $n_j(w) \neq 0$. Define the left- S -linear map $\psi: S^{(\mathbb{N})} \rightarrow E$ on the free basis by $\psi(e_j) = w$ and $\psi(e_i) = 0$ for $i \neq j$. Then $\psi(n) = \sum_i n_i \psi(e_i) = n_j \cdot w = n_j(w) \neq 0$, so $\psi|_N \in \text{Hom}_S(N, E)$ is nonzero. $\square$

Theorem 4.2 (Main Theorem). *If M is projective, then $2^{\aleph_0} = 2^{\aleph_1}$.*

Proof. Suppose M is projective.

Kaplansky decomposition. By Kaplansky's theorem [7] a projective module is a direct sum of countably generated submodules: $M = \bigoplus_{i \in I} M_i$ with each M_i countably generated projective; discard zero summands, so every $M_i \neq 0$.

The index set is uncountable. If $|I| \leq \aleph_0$ then M is a countable direct sum of countably generated modules, hence countably generated, contradicting Lemma 2.7. So $|I| \geq \aleph_1$.

Dual of the coproduct. Since $\text{Hom}_S(-, E)$ turns coproducts into products,

$$\text{Hom}_S(M, E) = \text{Hom}_S\left(\bigoplus_{i \in I} M_i, E\right) = \prod_{i \in I} \text{Hom}_S(M_i, E).$$

Lower bound. By Lemma 4.1 each $V_i := \text{Hom}_S(M_i, E) \neq 0$; pick $0 \neq v_i \in V_i$. Then $k^I \rightarrow \prod_i V_i$, $(a_i) \mapsto (a_i v_i)$, is k -linear and injective, so $\dim_k \text{Hom}_S(M, E) \geq \dim_k k^I$. By the

Erdős–Kaplansky theorem, for infinite I one has $\dim_k k^I = |k|^{|I|} \geq 2^{|I|}$. Hence, using $|I| \geq \aleph_1$ and monotonicity of the continuum function,

$$\dim_k \text{Hom}_S(M, E) \geq 2^{|I|} \geq 2^{\aleph_1}.$$

Upper bound. By Lemma 2.8, $\dim_k \text{Hom}_S(M, E) = 2^{\aleph_0}$.

Collision. Therefore $2^{\aleph_1} \leq \dim_k \text{Hom}_S(M, E) = 2^{\aleph_0} \leq 2^{\aleph_1}$, so $2^{\aleph_0} = 2^{\aleph_1}$. $\square$

Corollary 4.3. *If $2^{\aleph_0} < 2^{\aleph_1}$ —in particular under CH ($2^{\aleph_0} = \aleph_1 < \aleph_2 \leq 2^{\aleph_1}$) and under GCH—then M is not projective, so Vec is $\triangleleft$ -inadmissible and Conjecture 2.2 is false. Via Theorem 3.4, M is also not R -projective under CH.*

Corollary 4.4. *Conjecture 2.2, equivalently $\triangleleft$ -admissibility of Vec , is not a theorem of ZFC: it fails in every model of CH, and CH is consistent with ZFC.*

Remark 4.5 (The correction). The reduction note [9, (P2)] computed, from a hypothetical Kaplansky decomposition, exactly the inequality $2^{|I|} \leq 2^{\aleph_0}$ and concluded “cardinality cannot refute projectivity.” The missing step is $|I| \geq \aleph_1$ (Lemma 2.7), which turns $2^{|I|} \leq 2^{\aleph_0}$ into $2^{\aleph_1} \leq 2^{\aleph_0}$. The inequality is not weak: it asserts that the continuum function is constant on $[\aleph_0, \aleph_1]$, a demand that CH refutes.

Remark 4.6 (Why conditional, not absolute: the $\mathbb{Z}^{\mathbb{N}}$ parallel). The same schema proves the Baer–Specker group $\mathbb{Z}^{\mathbb{N}}$ non-free over $\mathbb{Z}$: a free decomposition $\mathbb{Z}^{\mathbb{N}} = \bigoplus_{i \in I} \mathbb{Z}$ has $|I| = \mathfrak{c}$, so $\text{Hom}(\mathbb{Z}^{\mathbb{N}}, \mathbb{Z}) = \prod_I \mathbb{Z}$ would have cardinality $2^{\mathfrak{c}}$, while Specker’s theorem gives $\text{Hom}(\mathbb{Z}^{\mathbb{N}}, \mathbb{Z}) = \mathbb{Z}^{(\mathbb{N})}$ of cardinality $\aleph_0$ —an *absolute* contradiction, because the dual collapses to countable rank. Over S , Theorem 2.5 keeps the dual large ($\dim_k \text{Hom}_S(M, E) = \mathfrak{c}$, Lemma 2.8), so the contradiction is only $2^{\aleph_1}$ versus $2^{\aleph_0}$. The “void Baer–Specker” of Corollary 2.6 is exactly what demotes an absolute ZFC refutation to a CH-conditional one: same count, different dual size, different verdict.

5. STRUCTURAL PLACEMENT AND THE PROJECTIVE DIMENSION OF M

The Main Theorem is silent when $2^{\aleph_0} = 2^{\aleph_1}$. To locate the residual question we do two things: place M in the almost-free hierarchy, and reframe the conjecture as the value of a single homological invariant, $\text{pd}_S M$, that does not name the continuum. The reframing turns admissibility into a recognisable classical problem—whether a product raises projective dimension—and yields the bounds that delimit the residual.

5.1. The almost-free profile.

Proposition 5.1 (M is flat, Mittag–Leffler, $\aleph_1$ -projective). *M is flat and Mittag–Leffler; hence $\aleph_1$ -projective. Consequently*

Conjecture 2.2 $\iff M$ is a direct sum of countably generated submodules.

Proof. Over the von Neumann regular S every module is flat and every submodule is pure (Proposition 2.4); $M = \prod_r P \subseteq \Phi = \prod_{r,m} E$ is a pure submodule of a product of copies of the finitely presented module E , so by the Raynaud–Gruson purity characterisation [15] M is Mittag–Leffler. Flat Mittag–Leffler modules are $\aleph_1$ -projective. For the equivalence: Kaplansky gives projective $\implies$ direct sum of countably generated; conversely a countably generated flat Mittag–Leffler module is projective [15, 18], so a direct sum of countably generated (necessarily flat ML) submodules of M is a direct sum of projectives, hence projective. $\square$

Thus M carries the exact structural profile of the Baer–Specker group $\mathbb{Z}^{\mathbb{N}}$: flat, Mittag–Leffler, $\aleph_1$ -projective, and (under CH) non-projective. Its non-projectivity is a *global* failure to decompose, orthogonal to the local projectivity it manifestly has.

5.2. The seed and the projective-dimension reframing. The next lemma names the small module out of which M 's defect is assembled.

Lemma 5.2 (The pd-1 seed). *Let $I \subseteq S$ be the left ideal of finite-column-support matrices. Then $I \cong P = \bigoplus_{\mathbb{N}} E$, and*

$$N_0 := S/I \cong \prod_{\mathbb{N}} E / \bigoplus_{\mathbb{N}} E$$

is cyclic, flat, and non-projective, with $\text{pd}_S N_0 = 1$.

Proof. Under the freeness isomorphism $\Psi: S \xrightarrow{\sim} \prod_{\mathbb{N}} E$, $s \mapsto (s \cdot e_m)_m$ of Theorem 2.5, the finite-support column sequences $\bigoplus_{\mathbb{N}} E$ correspond precisely to the finite-column-support matrices I ; hence $I \cong \bigoplus_{\mathbb{N}} E = P$ (projective, non-finitely-generated) and $S/I \cong \prod_{\mathbb{N}} E / \bigoplus_{\mathbb{N}} E$. Over the von Neumann regular S a finitely generated left ideal is a direct summand Se ; so S/I is projective iff I is a direct summand iff I is finitely generated. But I is not: any n elements have support in finitely many columns, hence lie in a proper subsum $\bigoplus_{m < m_0} E \subsetneq \bigoplus_{\mathbb{N}} E$. So S/I is non-projective, and $0 \rightarrow I \rightarrow S \rightarrow N_0 \rightarrow 0$ with $I \cong P$ projective and S free gives $\text{pd}_S N_0 = 1$. Flatness is automatic over S . $\square$

The seed is the scalar analogue of the Baer–Specker quotient $\mathbb{Z}^{\mathbb{N}}/\mathbb{Z}^{(\mathbb{N})}$; the field makes it *cyclic*, which $\mathbb{Z}$ does not. Products are exact, so taking $\prod_{\mathbb{N}}$ of the seed's resolution assembles M .

Lemma 5.3 (The controlling cokernel). $\Phi/M \cong \prod_{\mathbb{N}} N_0$, and the free presentation of this quotient is

$$0 \rightarrow M \rightarrow \Phi \cong S \rightarrow \prod_{\mathbb{N}} N_0 \rightarrow 0, \quad \Phi \cong S \text{ free.}$$

Proof. $\Phi = \prod_r (\prod_m E)$ and $M = \prod_r (\bigoplus_m E)$; products are exact, so $\Phi/M = \prod_r (\prod_m E / \bigoplus_m E) \cong \prod_r N_0$ by Lemma 5.2. The displayed sequence is the product over r of $0 \rightarrow I \rightarrow S \rightarrow N_0 \rightarrow 0$: indeed $\prod_r I = \prod_r P = M$ and $\prod_r S \cong S$ by Theorem 2.5. $\square$

Theorem 5.4 (Projective-dimension reframing, ZFC).

$$\text{Conjecture 2.2} \iff M \text{ projective} \iff \text{pd}_S M = 0 \iff \text{pd}_S \left(\prod_{\mathbb{N}} N_0 \right) \leq 1,$$

and $\text{pd}_S (\prod_{\mathbb{N}} N_0) = \text{pd}_S M + 1$ whenever M is non-projective.

Proof. From the free presentation $0 \rightarrow M \rightarrow S \rightarrow \prod_{\mathbb{N}} N_0 \rightarrow 0$ of Lemma 5.3 with S free, dimension shifting gives $\text{Ext}_S^{n+1}(\prod_{\mathbb{N}} N_0, -) \cong \text{Ext}_S^n(M, -)$ for all $n \geq 1$; and $\prod_{\mathbb{N}} N_0$ has projective dimension ≤ 1 exactly when the connecting map makes M projective. The first two equivalences are Theorem 2.3 and the definition of projective dimension. $\square$

The reading is the point. The seed N_0 has $\text{pd} = 1$; the controlling cokernel $\prod_{\mathbb{N}} N_0$ is a *countable product of copies of a pd-1 module*. Over a ring where projective dimension is not additive across products—any non-perfect ring, S included—such a product can climb to $\text{pd} = 2$, and whether it does is exactly Conjecture 2.2. This is the Osofsky–Jensen phenomenon that a product of modules can raise projective dimension by an amount sensitive to the continuum, now recognised as the precise content of the admissibility question.

5.3. Bounds on $\text{pd}_S M$. Write $d := \min\{n : 2^{\aleph_0} \leq \aleph_n\}$, so $\mathfrak{c} \leq \aleph_d$ and $d = 1$ iff CH.

Proposition 5.5 (Bounds, ZFC). *$1 \leq \text{pd}_S M \leq d + 1$ whenever $2^{\aleph_0} < 2^{\aleph_1}$, and $\text{pd}_S M \leq d + 1$ in every model.*

Proof. Lower bound: if $2^{\aleph_0} < 2^{\aleph_1}$ then M is non-projective (Theorem 4.2), i.e. $\text{pd}_S M \geq 1$. Upper bound: M is flat (Proposition 2.4) and generated by $|M| \leq \mathfrak{c} \leq \aleph_d$ elements, so the Gruson–Jensen estimate—an $\aleph_n$ -generated flat module has projective dimension $\leq n + 1$ [5]—gives $\text{pd}_S M \leq d + 1$. $\square$

Corollary 5.6 (Refined necessary condition, ZFC). *If M is projective then $2^{\aleph_0} = 2^{\aleph_1}$ and $M = \bigoplus_{i \in I} M_i$ with each $M_i \neq 0$ countably generated projective, $2^{|I|} \leq 2^{\aleph_0}$ and $|I| \in [\aleph_1, \mathfrak{c}]$. In the natural residual model $\text{MA} + 2^{\aleph_0} = \aleph_2$ (where $2^{\aleph_1} = \aleph_2 < 2^{\aleph_2}$) the rank $|I|$ is forced to be exactly $\aleph_1$.*

Proof. Kaplansky decomposes projective $M = \bigoplus_{i \in I} M_i$; Lemma 2.7 gives $|I| \geq \aleph_1$, and the dual count of Theorem 4.2 (Lemma 2.8, $\dim_k \text{Hom}_S(M, E) = 2^{\aleph_0}$, against $\dim_k k^I \geq 2^{|I|}$) gives $2^{|I|} \leq 2^{\aleph_0}$, whence $2^{\aleph_0} = 2^{\aleph_1}$. As each $|M_i| \leq \mathfrak{c}$, $|M| = \mathfrak{c}$ bounds $|I| \leq \mathfrak{c}$. Under $\text{MA} + 2^{\aleph_0} = \aleph_2$ the only cardinal $\lambda \in [\aleph_1, \aleph_2]$ with $2^\lambda \leq 2^{\aleph_0} = \aleph_2$ is $\lambda = \aleph_1$ (since $2^{\aleph_2} > \aleph_2$), so $|I| = \aleph_1$. $\square$

So in the very model where independence could live, the hypothetical decomposition is $\aleph_1$ -indexed—an ω_1 -scale object, precisely the size at which the almost-free/ Γ -invariant machinery operates, even though M itself is $\mathfrak{c}$ -sized and its projective dimension is continuum-governed.

Remark 5.7 (An unverified sharpening). Osofsky’s continuum-sensitive computation of the global dimension of $\text{End}_k(E)$ [14] would, on the value $\text{l.gl.dim } S = d + 1$, sharpen the upper bound: as $M \subseteq \Phi$ is a submodule of a free module, $\text{pd}_S M = \text{pd}_S(\prod_{\mathbb{N}} N_0) - 1 \leq \text{gl.dim } S - 1 = d$, giving $\text{pd}_S M = 1$ exactly under CH. We flag this as resting on a recollection of Osofsky’s exact value that we have *not* verified at source; no result below depends on it, and the ZFC-clean bound of Proposition 5.5 ($\text{pd}_S M \in \{1, 2\}$ under CH) stands unconditionally.

5.4. An order-free obstruction: κ -projectivity. The cardinal count of Section 4 is capped: its gap $2^{\aleph_1}$ -versus- $2^{\aleph_0}$ is vacuous when $2^{\aleph_0} = 2^{\aleph_1}$. We record one continuum-independent obstruction whose failure, if provable, would decide the conjecture in ZFC.

Definition 5.8. For κ a regular uncountable cardinal, a module N is κ -projective if it has a κ -dense system of $< \kappa$ -generated projective submodules: a set $\mathcal{D}$ of $< \kappa$ -generated projective submodules such that every $< \kappa$ -generated submodule of N lies in some member of $\mathcal{D}$, and $\mathcal{D}$ is closed under unions of well-ordered ascending chains of length $< \kappa$.

Lemma 5.9 (κ -projectivity reduction, ZFC). *If M is projective, then M is κ -projective for every regular uncountable κ . Consequently, if M fails to be κ -projective for some such κ , then Conjecture 2.2 is false and Vec is $\triangleleft$ -inadmissible in ZFC.*

Proof. By Kaplansky $M = \bigoplus_{i \in I} M_i$ with each M_i countably generated projective. Put $\mathcal{D} = \{\bigoplus_{i \in J} M_i : J \subseteq I, |J| < \kappa\}$. Each member is projective and, as κ is regular uncountable, $< \kappa$ -generated. Density: a $< \kappa$ -generated $N \subseteq M$ has generators of finite support in I , so their supports union to $J \subseteq I$ with $|J| < \kappa$ (regularity) and $N \subseteq \bigoplus_{i \in J} M_i \in \mathcal{D}$. Chain-closure: a union of $< \kappa$ members is $\bigoplus_{i \in J_\xi} M_i$ with $|\bigcup J_\xi| < \kappa$, again in $\mathcal{D}$. The contrapositive is the stated consequence. $\square$

Since $\aleph_1$ -projectivity holds automatically (Proposition 5.1), the first substantive κ -projectivity test is $\kappa = \aleph_2$; a ZFC failure there is one route to a ZFC refutation. Both this test and the projective-dimension reframing point at the same residual, which we now state as a single question.

Question 5.10. Compute $\text{pd}_S M$ as a function of the continuum. Concretely: can $\text{pd}_S M = 0$ in a model of $2^{\aleph_0} = 2^{\aleph_1}$? A ZFC proof that $\text{pd}_S M \geq 1$ in every model makes Conjecture 2.2 *false in ZFC*; a construction of a countable-sum decomposition (equivalently, $\text{pd}_S M = 0$) under $\text{MA} + \neg\text{CH}$ makes it *independent*. The $\aleph_2$ -projectivity test of Lemma 5.9 is the $\kappa = \aleph_2$ facet of this question.

6. THE RESIDUAL CASE AND THE INDEPENDENCE LEAN

By Theorem 5.4 the residual is a single invariant: *can $\text{pd}_S M = 0$ in a model of $2^{\aleph_0} = 2^{\aleph_1}$?* We now weigh the evidence on both sides. It is genuinely two-sided: both a ZFC refutation and full independence remain live, and we do not claim either. The results of this section that quote the Herbera–Trlifaj almost-free theory [6] situate the problem; the section’s proved content is entirely in Section 5.

The ambient class over S (verified). Herbera and Trlifaj’s Example 6.8 [6] is stated for “a non-artinian von Neumann regular right self-injective ring, for example the endomorphism ring of an infinite dimensional [linear] space”—this is S verbatim. We have verified the relevant statements at source. Theorem 2.9(i) of [6] proves, over any ring in ZFC, that a module is $\aleph_1$ -projective iff it is flat Mittag–Leffler; Example 6.8 shows that over S these coincide further with the non-singular modules (via S right non-singular and Goodearl’s characterisation of finitely generated non-singular submodules), so the identification $\text{flat ML} = \aleph_1\text{-projective} = \text{non-singular}$ is ZFC-absolute; and Corollary 7.3 states that the class of flat Mittag–Leffler modules is deconstructible iff the ring is right perfect, so over the non-perfect S it is *not* deconstructible (it is a Kaplansky class with bound $2^{|S|} = 2^{\aleph}$). Hence M ’s membership in the ambient class is cheap and absolute (M is non-singular), and Conjecture 2.2 asks whether this particular non-singular module is a direct sum of countably generated pieces—equivalently, $\text{pd}_S M = 0$ (Theorem 5.4).

Moreover [6, §5] constructs, *in ZFC*, an $\aleph_1$ -generated $\aleph_1$ -projective non-projective S -module W : from a countably generated flat non-projective module (our seed N_0 , or any such, exists because S is not right perfect) one forms a subquotient along a ladder system on a stationary set of $\aleph_0$ -cofinal ordinals. So “flat Mittag–Leffler $\Rightarrow$ projective” is already false over S in ZFC, and it fails at the ω_1 scale. But W is a bespoke subquotient carrying an engineered stationary ladder, whereas $M = \prod_r P$ is a canonical product carrying no such ladder; whether M contains a non-absorbable copy of such a W is the crux, and it is exactly the question of whether $\text{pd}_S M \geq 1$ persists into the residual window.

Two intuitions pull opposite ways. We record both, at their honest weight.

Toward false in ZFC. M is a canonical continuum-sized product—precisely the shape of Osofsky’s high-projective-dimension witnesses. If $\text{pd}_S M$ tracks the continuum-degree $d \geq 1$ (as Osofsky’s global-dimension witnesses do), then M is non-projective in *every* model, and the counting’s “ $2^{\aleph_0} = 2^{\aleph_1}$ is necessary” is merely a weak necessary condition, never sufficient. Reinforcing this: the failure of “flat ML $\Rightarrow$ projective” witnessed by W is a size- $\aleph_1$ phenomenon living at ω_1 , where the seed and the ladder are all constructed in ZFC; the residual window $2^{\aleph_0} = 2^{\aleph_1}$ constrains the continuum function but does not obviously protect ω_1 . This is why we do *not* assert that “both obstructions vanish” in the residual window: the ω_1 -scale obstruction is not visibly demoted by a collapse of the continuum function.

Toward independence. The operative threshold in the count is the *collapse* $2^{\aleph_0} = 2^{\aleph_1}$ —an equalisation of the continuum function at $\aleph_1$, not the crude Osofsky degree $d = \min\{n : 2^{\aleph_0} \leq \aleph_n\}$ (which ignores $2^{\aleph_1}$). A collapse $2^{\aleph_0} = 2^{\aleph_1}$ is exactly what forcing arranges while preserving reflection principles; and $\text{pd}_S M = 0$ is, by Corollary 5.6, a Whitehead/ Γ -invariant statement about an $\aleph_1$ -indexed decomposition—the archetype of an independent assertion. The

disanalogy with $\mathbb{Z}^{\aleph}$, whose non-freeness is ZFC-*absolute* through slenderness, is precisely our *large dual* ($\prod_r E \cong S$ free, $\dim_k \text{Hom}_S(M, E) = \mathfrak{c}$; Theorem 2.5, Remark 4.6), which removes the absolute obstruction and leaves a reflection-sensitive one.

Remark 6.1 (Verdict). The exact value of $\text{pd}_S M$ depends on the continuum function *around* $\aleph_1$ —the $2^{\aleph_0}$ -versus- $2^{\aleph_1}$ threshold—rather than merely on the degree d , and that is the fingerprint of an independent, not absolute, invariant. On this basis the evidence *leans toward independence*: it is plausible that under $\text{MA} + 2^{\aleph_0} = \aleph_2$ the module M is a direct sum of countably generated submodules while under CH it is not, making Conjecture 2.2 independent of ZFC. But this is an evidential lean, *not a proof*, and false-in-ZFC is a genuine competitor: if $\text{pd}_S M$ tracks the continuum-degree the way Osofsky’s global witnesses do, M is non-projective in every model. Deciding between the two is exactly Question 5.10: an Osofsky-style computation of $\text{pd}_S M$ as a function of the continuum, most tractably attacked through the commutative shadow $\prod_r (k^\omega / k^{(\omega)})$ over k^ω .

7. CONSEQUENCES FOR THE CONTAINER PROGRAM

The composition of directed containers over Vec is not unconditionally well defined. In every model of the Continuum Hypothesis, the base Vec_k (over a countable field) fails to be $\triangleleft$ -admissible: there are containers whose substitution has no denotational meaning. Whether this failure is absolute—a theorem of ZFC—or merely conditional is the one remaining question, and the structural evidence leans toward the latter, which would make $\triangleleft$ -admissibility of Vec genuinely *independent* of ZFC; a ZFC refutation, however, remains live (Section 6).

Three points bear on the wider program.

First, this is the program’s first set-theoretically sensitive node. Every prior obstruction—the Zappa–Szép cocycle $[\omega] \in H^2$, the prunability descent class, the direction-functor cohomology—was a concrete, absolute invariant. Admissibility of the linear base is not: it lives in the cardinal arithmetic of the continuum. Any claim that “the theory works over Vec ” must now carry a set-theoretic hypothesis, or route around admissibility altogether (for instance by fixing a canonical convention for $p \triangleleft q$ rather than demanding closure of the image of $\llbracket - \rrbracket$).

Second, the negative result sharpens the value of the positive theory over Set and over the extensive bases, where admissibility is automatic and the composition is unconditional. The boundary between the two is now drawn exactly: admissibility has teeth precisely between the two degeneracies (the terminal unit of Set and the zero unit of Vec), and at the Vec end it is CH-sensitive.

Third, the module-theoretic core is of independent interest. The module $M = \prod_{\mathbb{N}} (\bigoplus_{\mathbb{N}} E)$ over the full linear ring is a natural, canonical example of a flat Mittag–Leffler $\aleph_1$ -projective module whose projectivity is CH-sensitive and possibly independent—a product-shaped, non-semiartinian companion to the Baer–Specker group, outside the reach of the packaged Dual-Baer independence theorems. Its projectivity is a single homological invariant: Conjecture 2.2 holds iff $\text{pd}_S M = 0$, and the controlling cokernel Φ/M is the countable product $\prod_{\mathbb{N}} N_0$ of a projective-dimension-one seed, so admissibility of Vec is an instance of the Osofsky–Jensen problem of whether a product raises projective dimension. The sharp open problem (Question 5.10) is to compute $\text{pd}_S M$ as a function of the continuum: a ZFC proof that $\text{pd}_S M \geq 1$ in every model collapses the question to FALSE-in-ZFC, a decomposition under $\text{MA} + \neg\text{CH}$ confirms independence, and either resolution would be the first theorem of its kind about the composition of directed containers.

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